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Cross-border Movement of $PM_{2.5}$ In Nepal

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Introduction

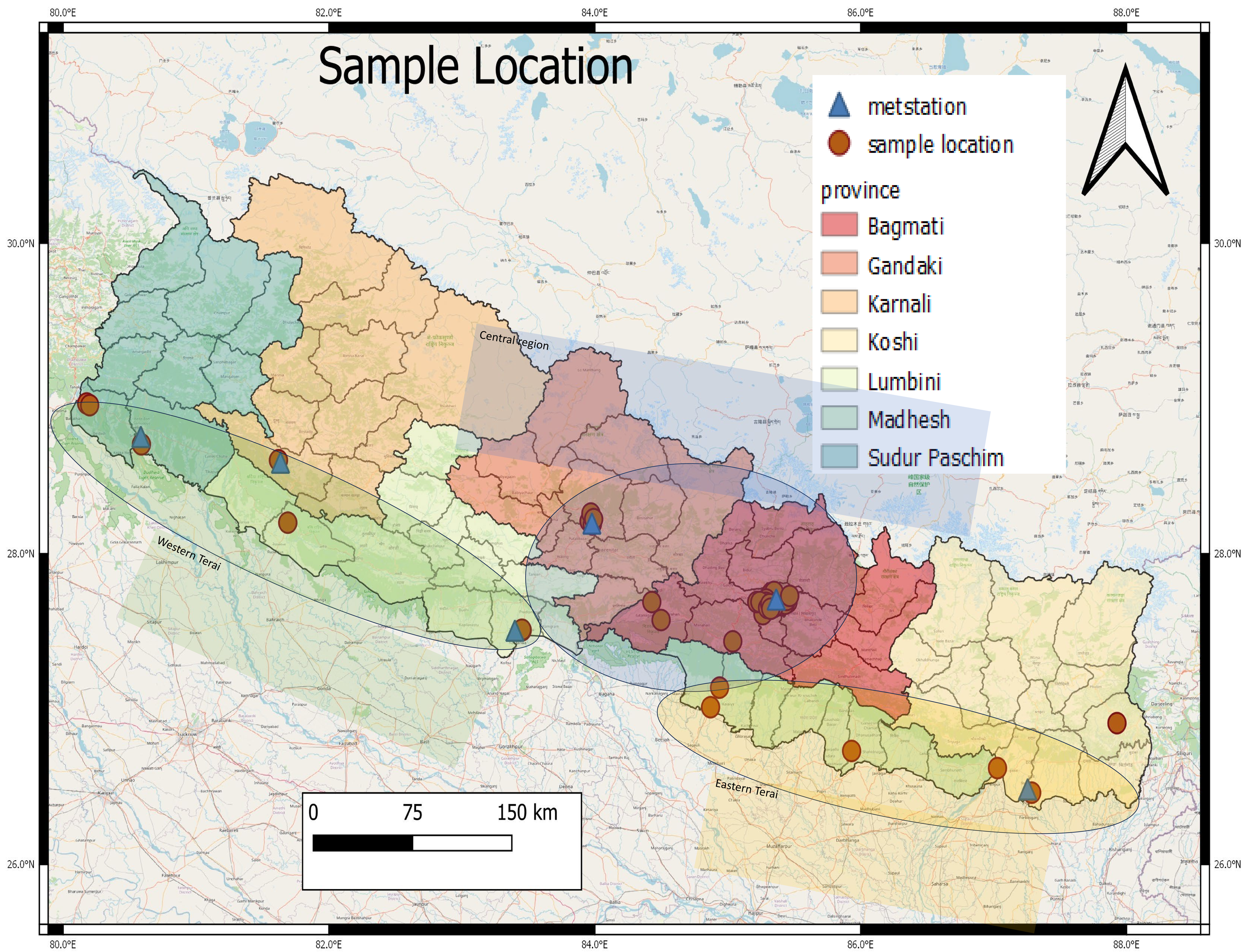
- Transboundary properties of air pollution contribute to the complexity air pollution and presents significant challenges for countries (Barrett, 1990).

- Indo-Gangetic Plains (IGP), is widely regarded as a hotspot for anthropogenic aerosol emission in South Asia. The region grapples with escalating levels of fine particulate matter, commonly referred to as $PM_{2.5}$ (particulate matter with a diameter of 2.5 micrometers or smaller), which not only poses a substantial threat to the health and well-being of its inhabitants but also challenges the very notion of transboundary environmental stewardship.

- Transboundary pollution in the Terai region is not solely a local issue; it epitomizes the far-reaching consequences of air quality deterioration, demanding a comprehensive understanding of the sources, transport mechanisms, and impacts of PM2.5 in a region.

- This manuscript aims to investigate the transboundary movement of $PM_{2.5}$ in the Terai region of Nepal, shedding light on the contributing factors, percentage of regional contribution, monitoring methods, and the implications for the environment and public health.

Sensor Locations



Sensors were installed in Western Terai, Eastern Terai and Central region.

In Terai, sensors were installed in pairs at 4 location to find the contribution of regional pollution.

Criteria for urban and rural pair:

- Location of sensor should be of same elevation.
- Urban and rural location should be in the radar of 50 km.
- Rural area should be the up wind of urban area so that there is no impact of urban pollution source in rural area.

Datasets

- $PM_{2.5}$ measurements from TSI Bluesky monitors
- HYSPLIT Back Trajectory: NCEP-GFS025
- GEOS Chem
- Windspeed and wind direction: Meteorological stations and ERA5

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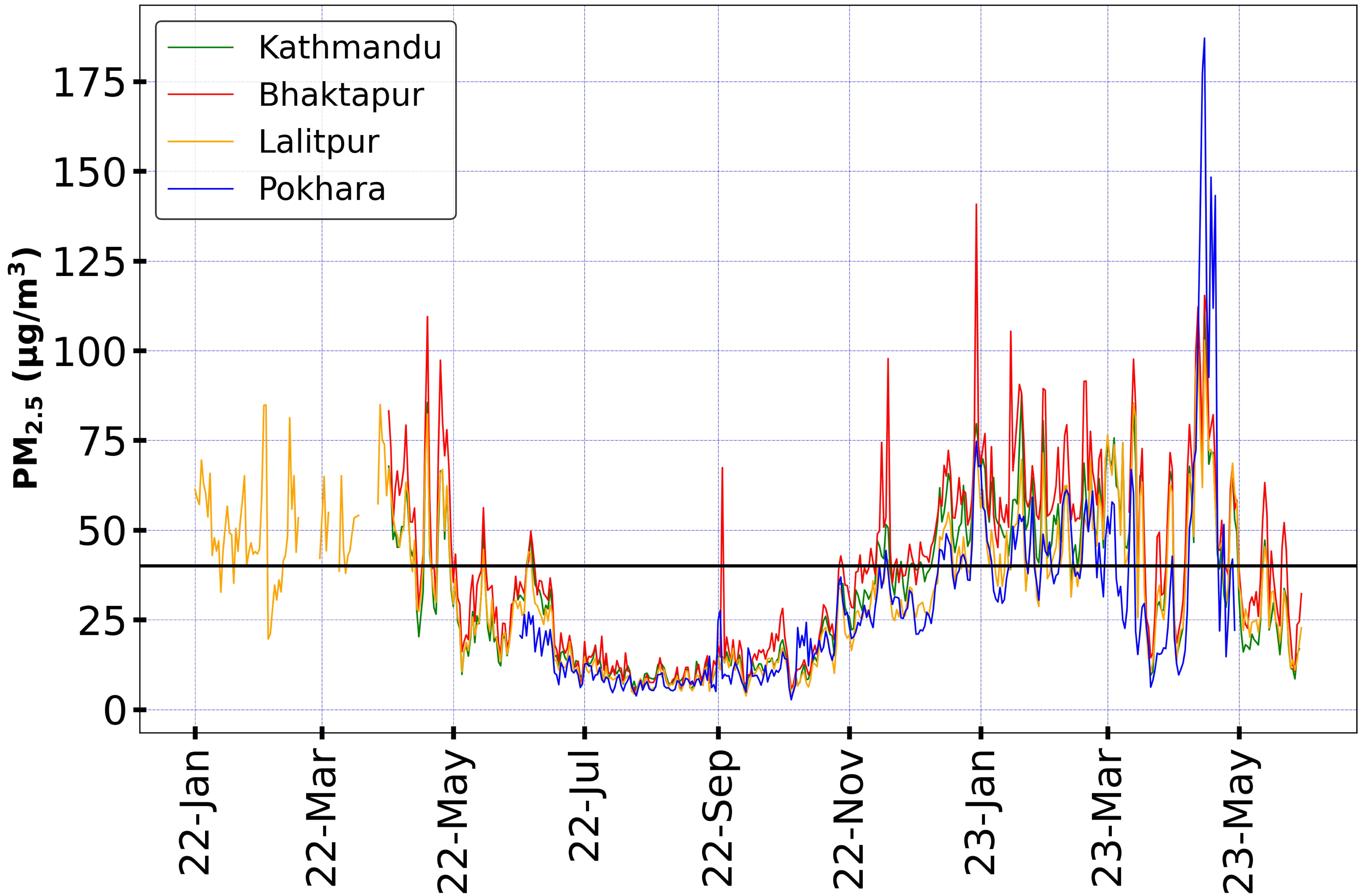
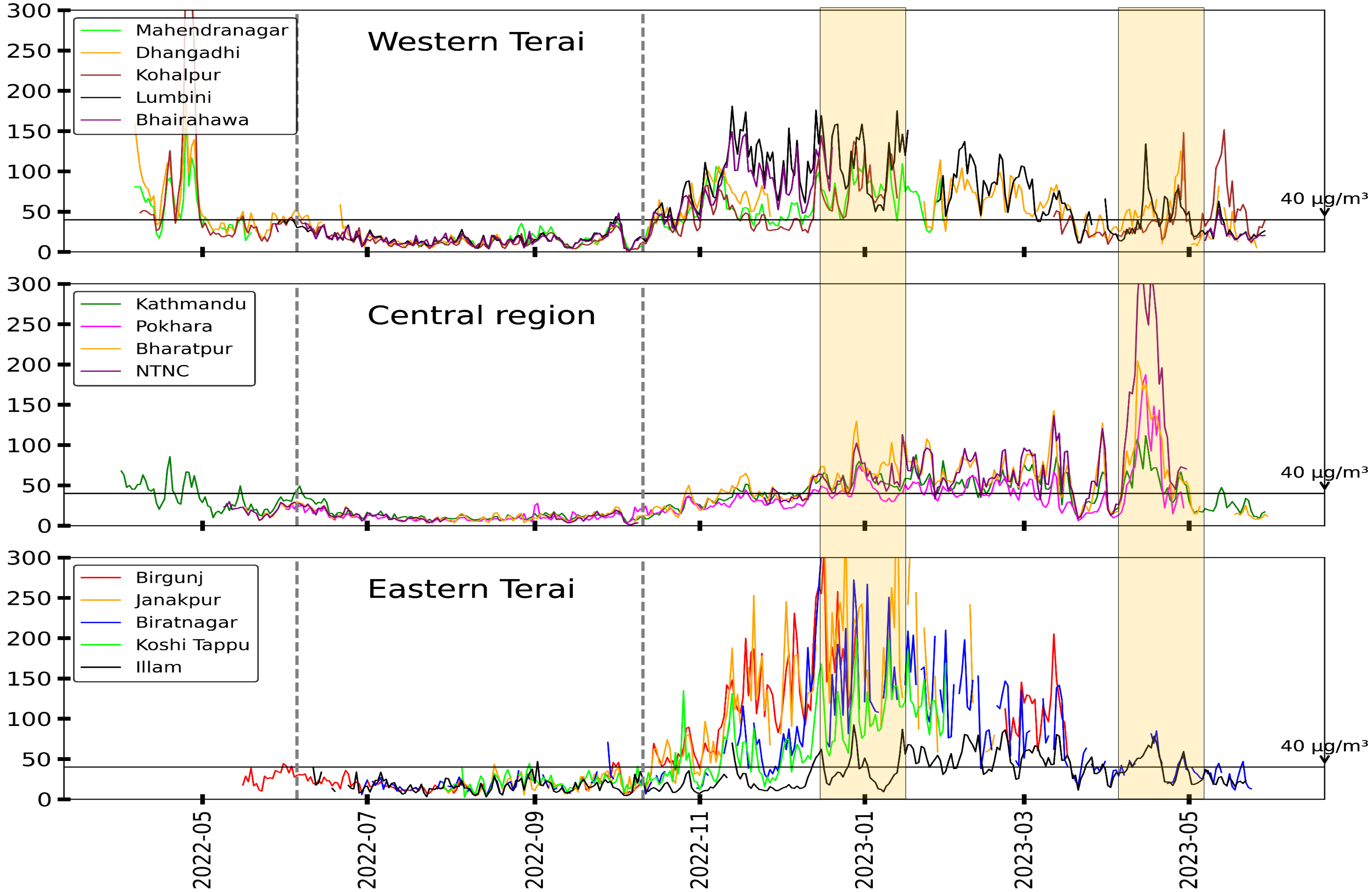
Variation of $PM_{2.5}$



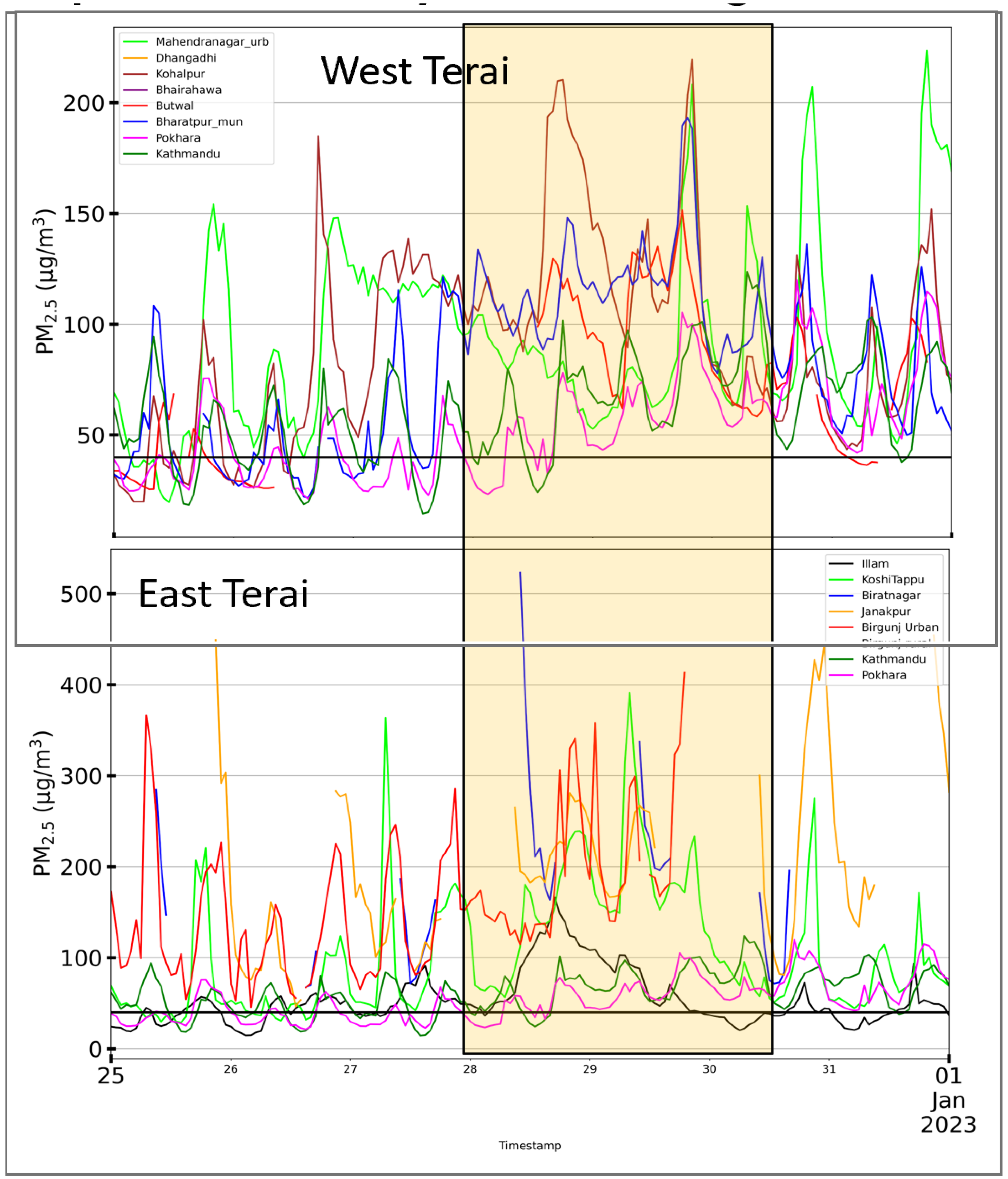
The time series of *average value* of sensor data were plotted. From it, three different timespan was selected, when $PM_{2.5}$ had increased abruptly in all considered locations.

Comparative Analysis



We compared the timeseries of central region with location in western and eastern Terai to see if there is similar pattern of rise and fall of $PM_{2.5}$ in all the region

Comparative Analysis

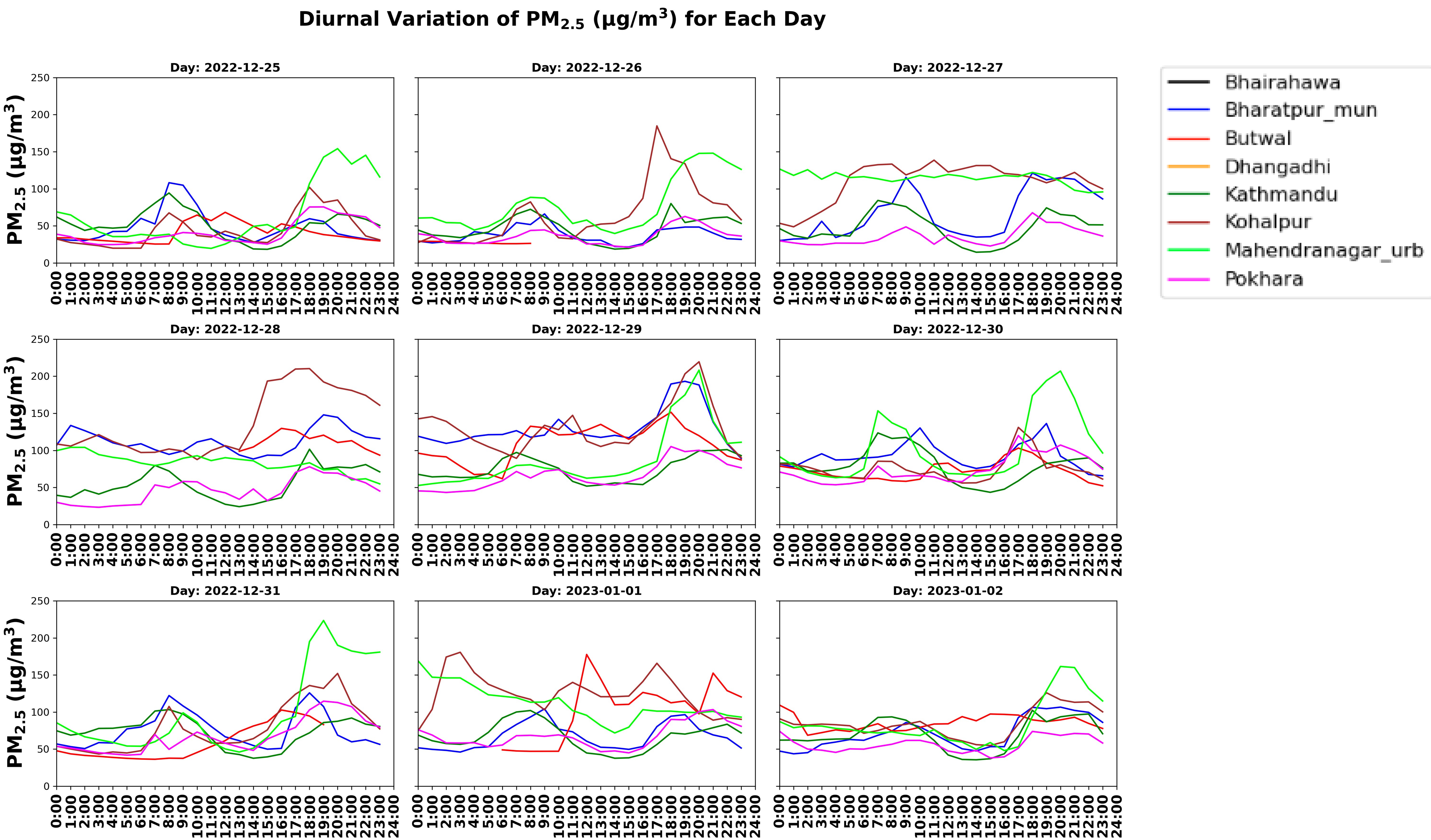


West to East

Date	Mahendranagar urban	Kohalpur	Butwal	Bharatpur urban	Pokhara	Kathmandu	Birgunj urban	Janakpur	Biratnagar	Koshi Tappu	Illam
12/25/2022	63.8	43.9	39.3	48.2	41.2	51.5	151.7	348.2	212.0	66.8	35.7
12/26/2022	78.4	65.1	27.3	37.4	36.1	45.5	113.9	126.8	81.3	57.1	41.2
12/27/2022	**114.3**	**109.5**		67.6	36.0	47.2	141.5	139.7	**122.4**	**110.4**	53.8
12/28/2022	**83.3**	**137.2**	**112.0**	**112.2**	45.8	54.5	**176.2**	**224.4**	**272.2**	**136.1**	**92.3**
12/29/2022	**88.5**	**130.8**	**109.4**	**129.5**	**66.6**	**74.5**	**227.4**	**214.3**	**231.3**	**199.7**	72.2
12/30/2022	**103.6**	75.5	72.2	**92.4**	**74.7**	**79.7**		**237.4**	**112.2**	96.2	37.8
12/31/2022	**99.1**	79.4	60.0	75.0	**67.7**	**73.2**		**242.4**		75.7	40.1
1/1/2023	**111.6**	125.0	105.7	66.9	**68.0**	**67.1**		146.0	121.0	80.0	51.1
1/2/2023	84.9	84.9	85.8	71.6	**56.3**	**69.8**		160.7	266.8	133.2	42.8
1/3/2023	88.2	108.1	65.1	63.8	**55.3**	**67.0**		92.8	122.4	90.3	26.9
1/4/2023	82.8	64.4	61.6	64.3	47.0	56.3		92.5	115.7	93.4	23.8
1/5/2023	84.3	62.7	50.2	55.6	31.5	55.6				166.6	16.9

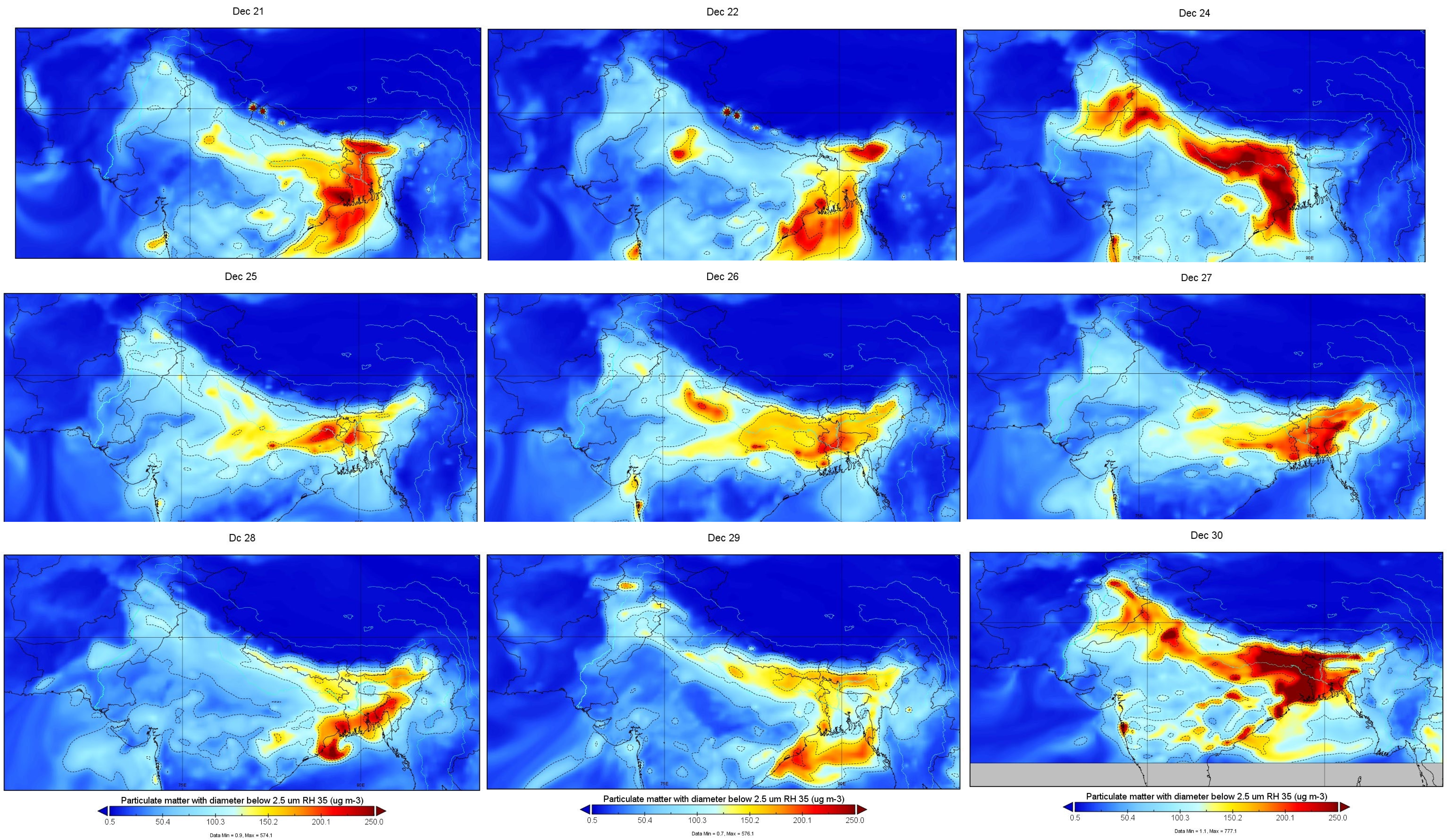
Comparison of daily 24-h average at different location at high $PM_{2.5}$ events in December 2022

Comparative Analysis



High pollution event also impacted the morning and evening peak

Variability of $PM_{2.5}$



Variability of $PM_{2.5}$ at Hindukush region derived from GEOS chem

Cluster Analysis

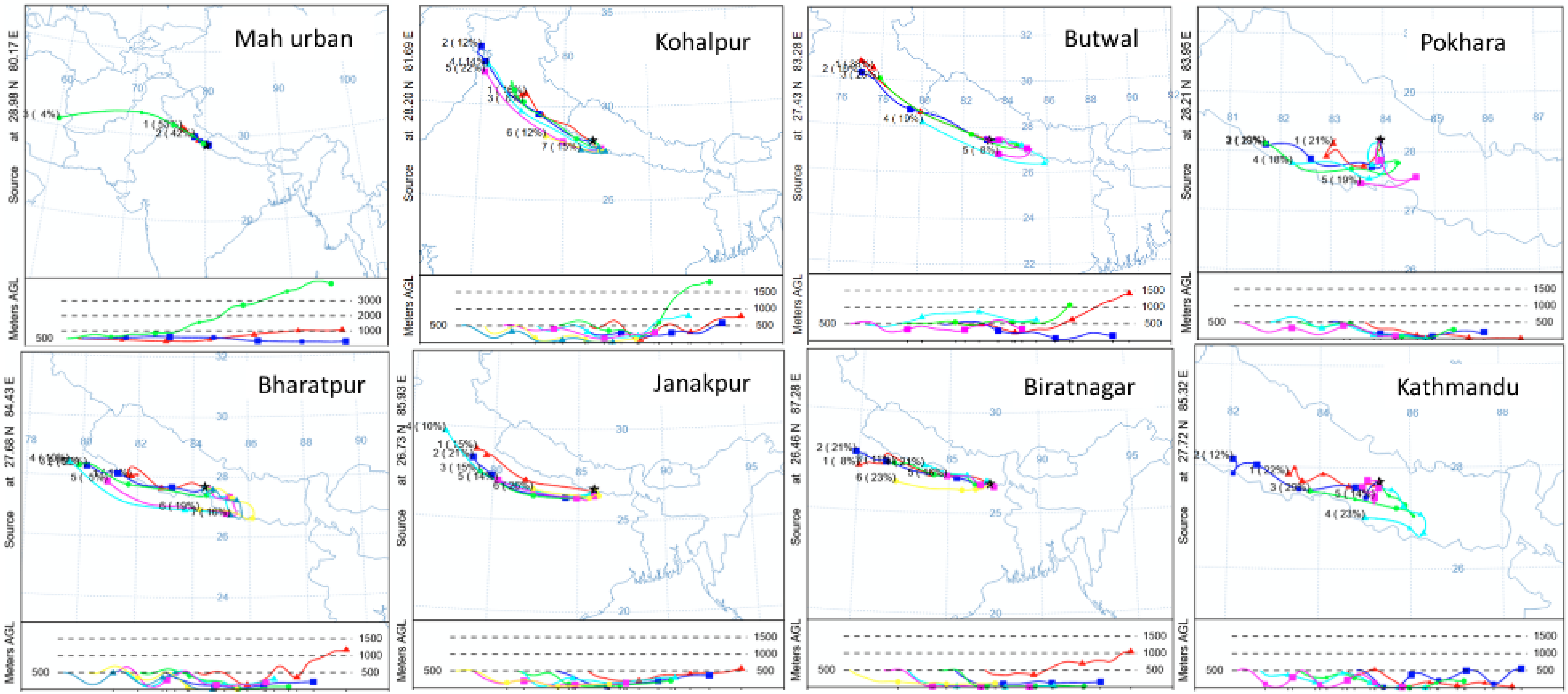


Figure 8: 3-days back trajectory clusters at different regions starting from 0:00 December 28 2022 (UTC).

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Back Trajectories

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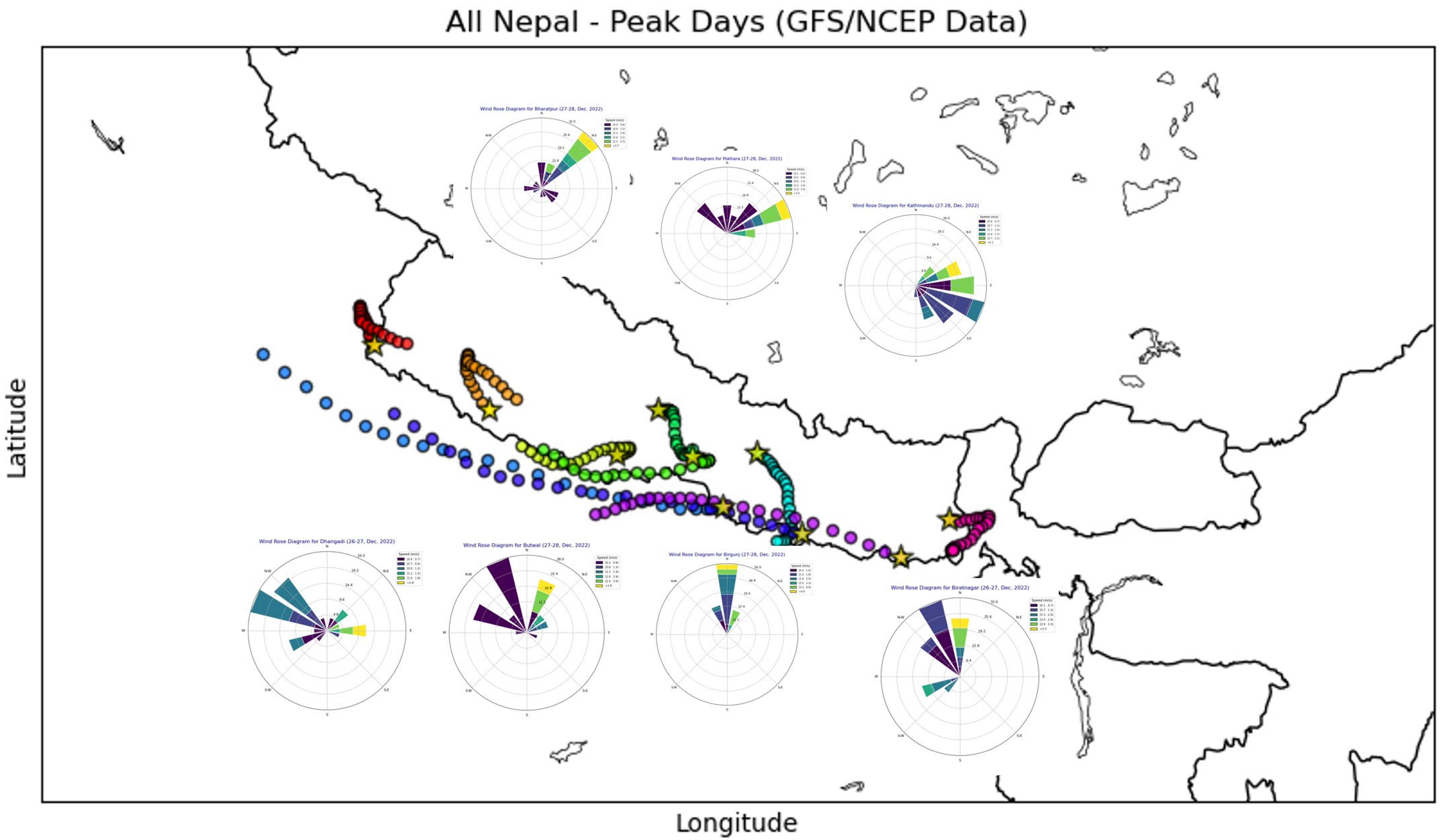
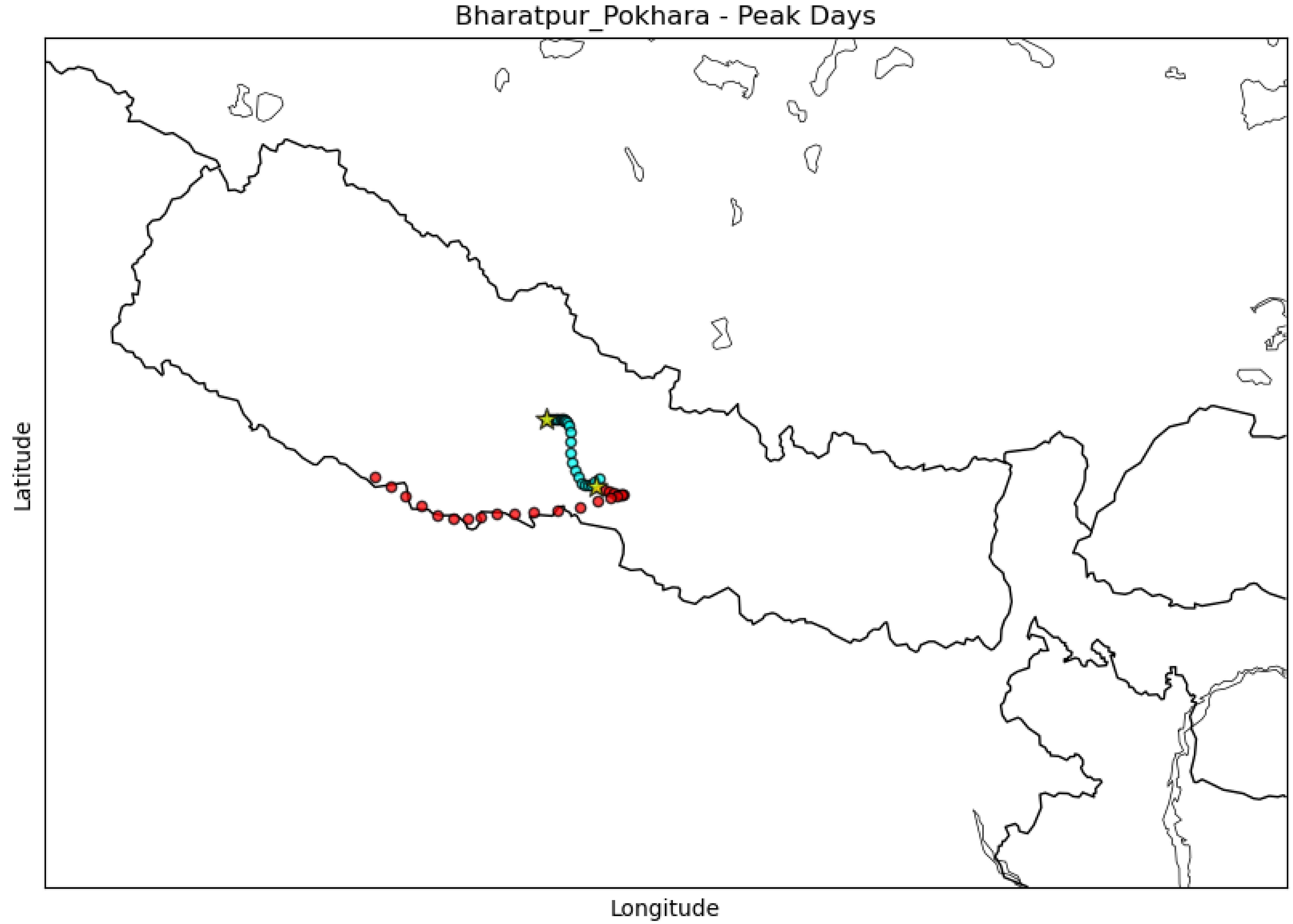


Case Study: I

Bharatpur
and
Pokhara

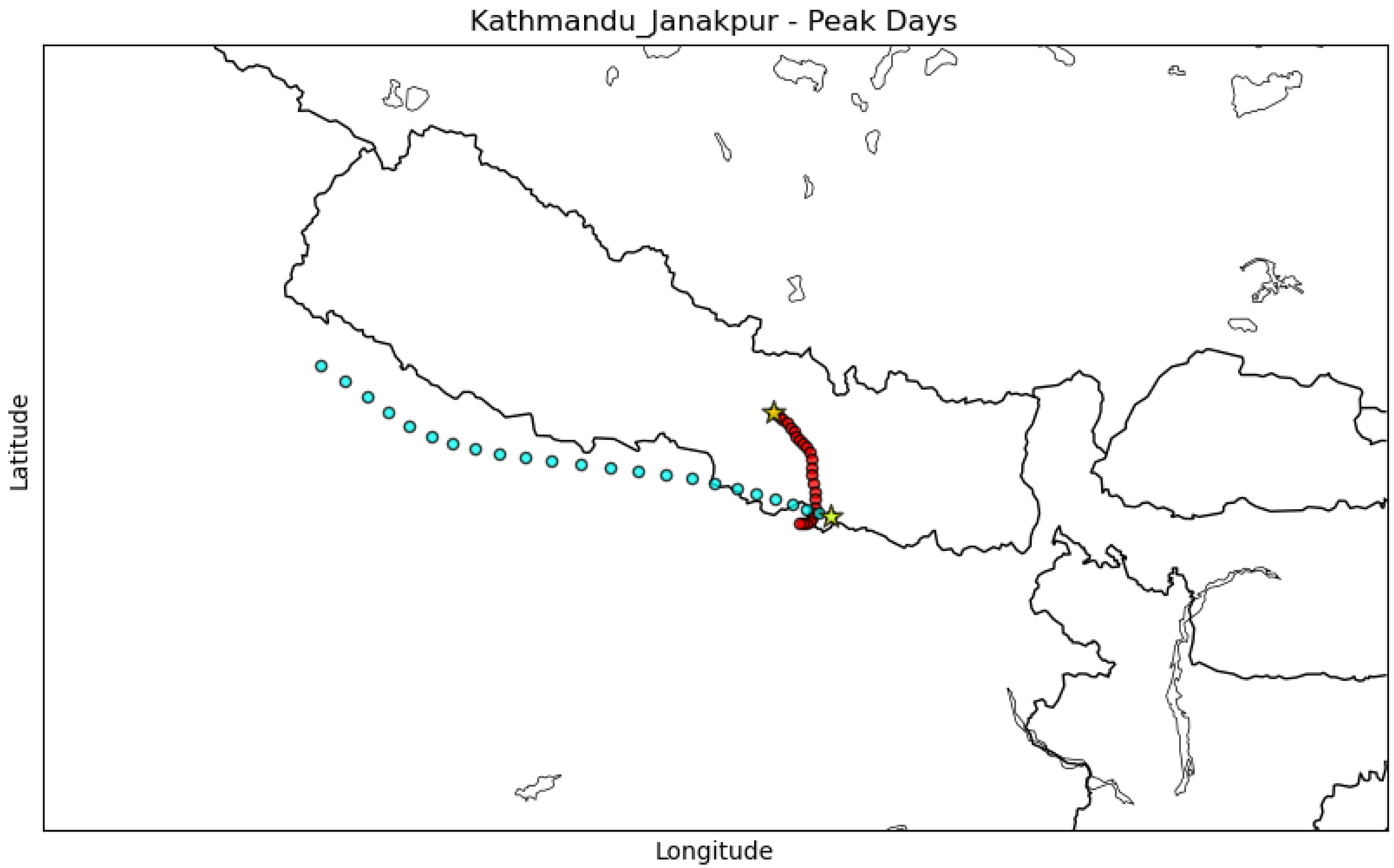


Case Study: II

Kathmandu

and

Janakpur



Contributions: *Local/Non-Local*

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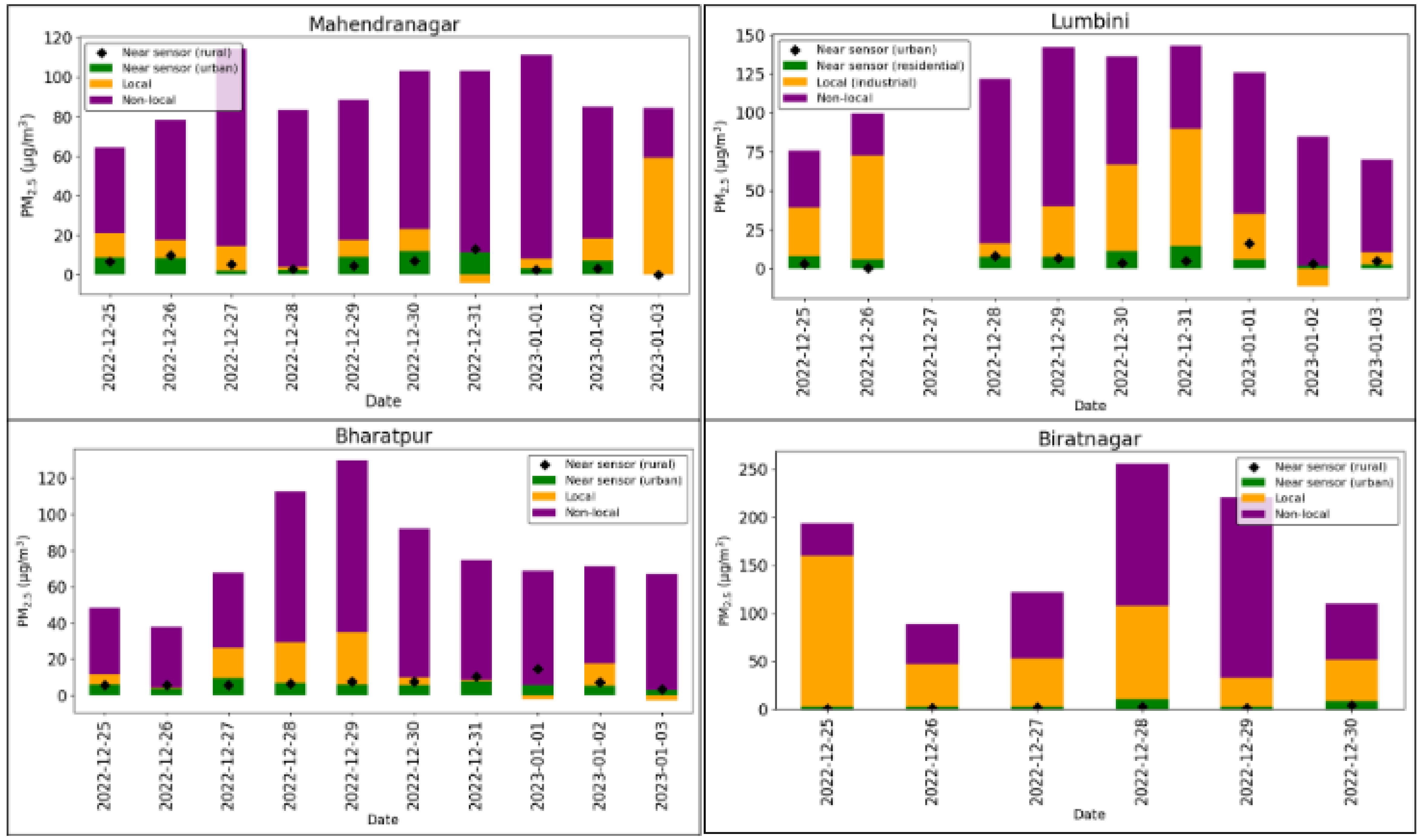

Figure 3: Contribution from rural near sensor, urban near sensor, local and non-local sources at different urban location of Terai using 1-hour moving average.

Timestamp	Mahendranagar			Lumbini			Bharatpur			Biratnagar		
	near sensor	local	non local	near sensor	local	non local	near sensor	local	non local	near sensor	local	non local
12/25/2022	13	19	67	11	41	48	13	11	76	2	81	18
12/26/2022	10	12	78	6	67	27	11	1	88	4	49	47
12/27/2022	2	10	88				15	25	61	3	41	56
12/28/2022	3	1	**96**	6	7	**87**	6	20	**74**	4	38	**58**
12/29/2022	10	10	**80**	5	23	**72**	5	22	**73**	1	14	**85**
12/30/2022	11	11	78	8	41	51	6	4	89	8	39	53
12/31/2022	11	-4	**93**	10	53	37	10	1	89			
1/1/2023	3	4	**93**	5	23	72	9	-3	**94**			
1/2/2023	8	13	78	2	-16	**113**	8	18	**75**			
12/3/2023		**70**	30	4	11	85	5	-4	**99**			

Local Scale Dynamics

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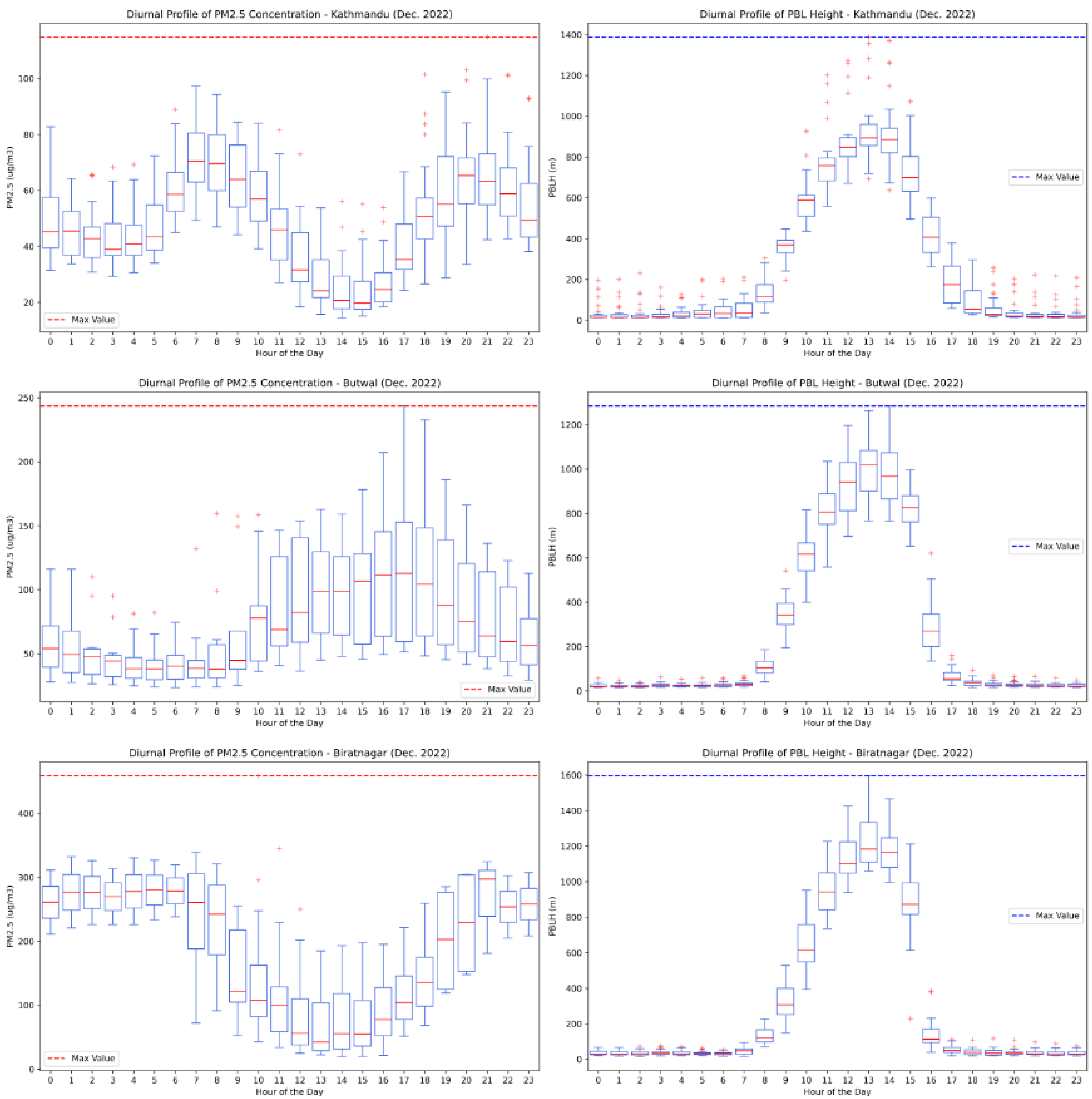

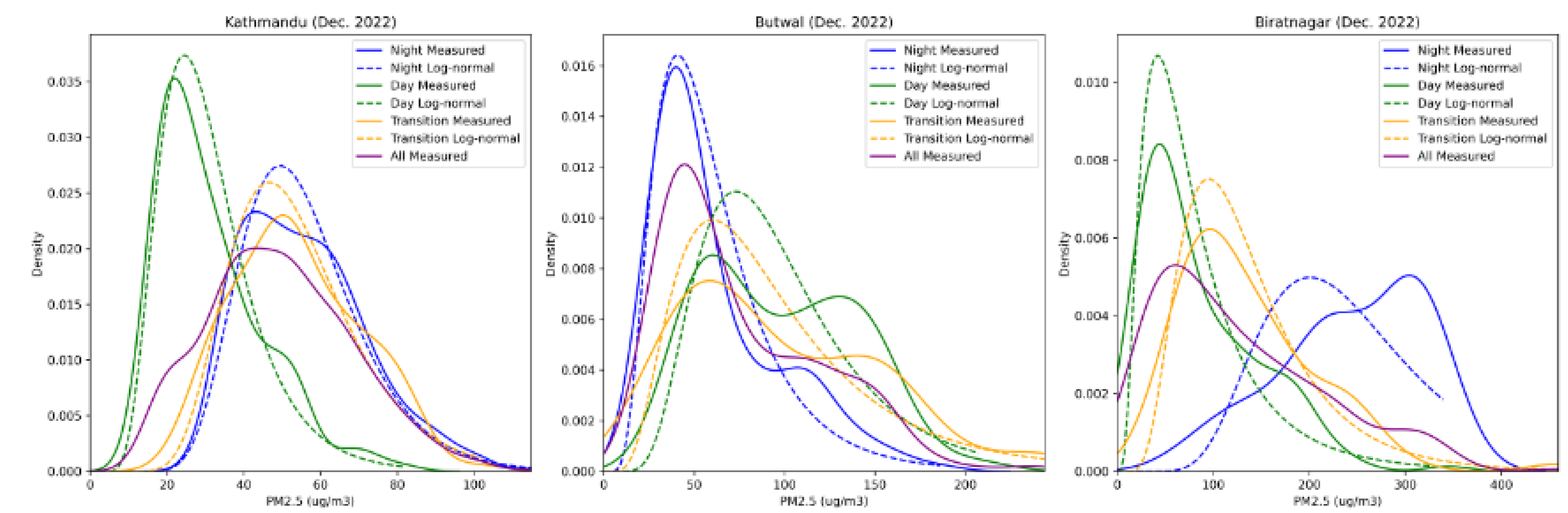

Figure 5: Frequency distribution of hourly $PM_{2.5}$ concentrations for Kathmandu, Butwal, and Biratnagar by time of day for December 2022. Night is defined as 19:00 to 8:00, Day is 11:00 to 16:00 and Transition includes 9:00, 10:00, 17:00 and 18:00. Solid lines show distribution of measurements, dashed lines show ideal log-normal distribution for comparison.

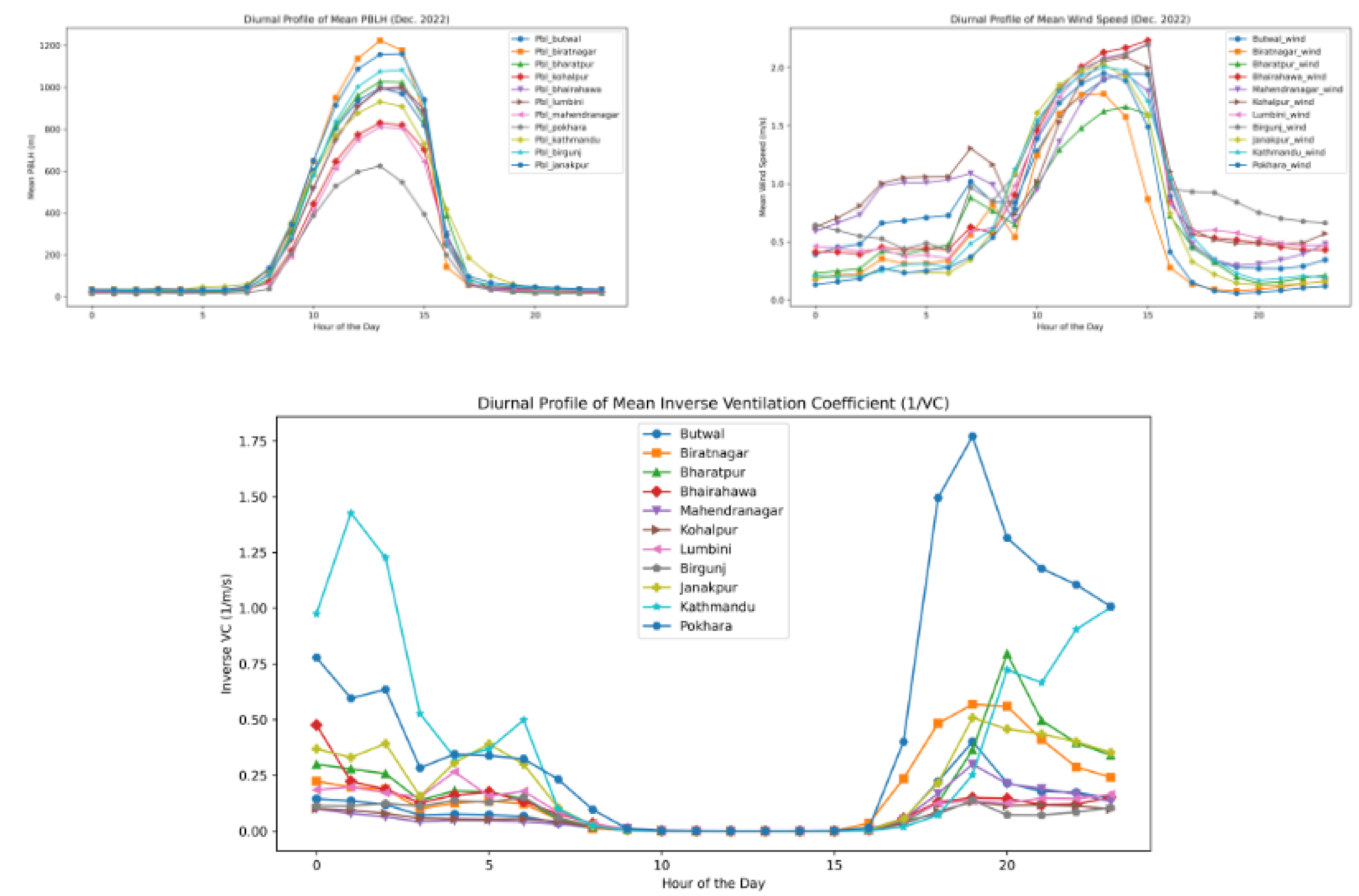

Figure 6: Diurnal profiles of PBLH (left), windspeed (right), and inverse ventilation coefficient (bottom) for all sensor locations in Nepal.

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Synoptic Scale Dynamics: PSCF

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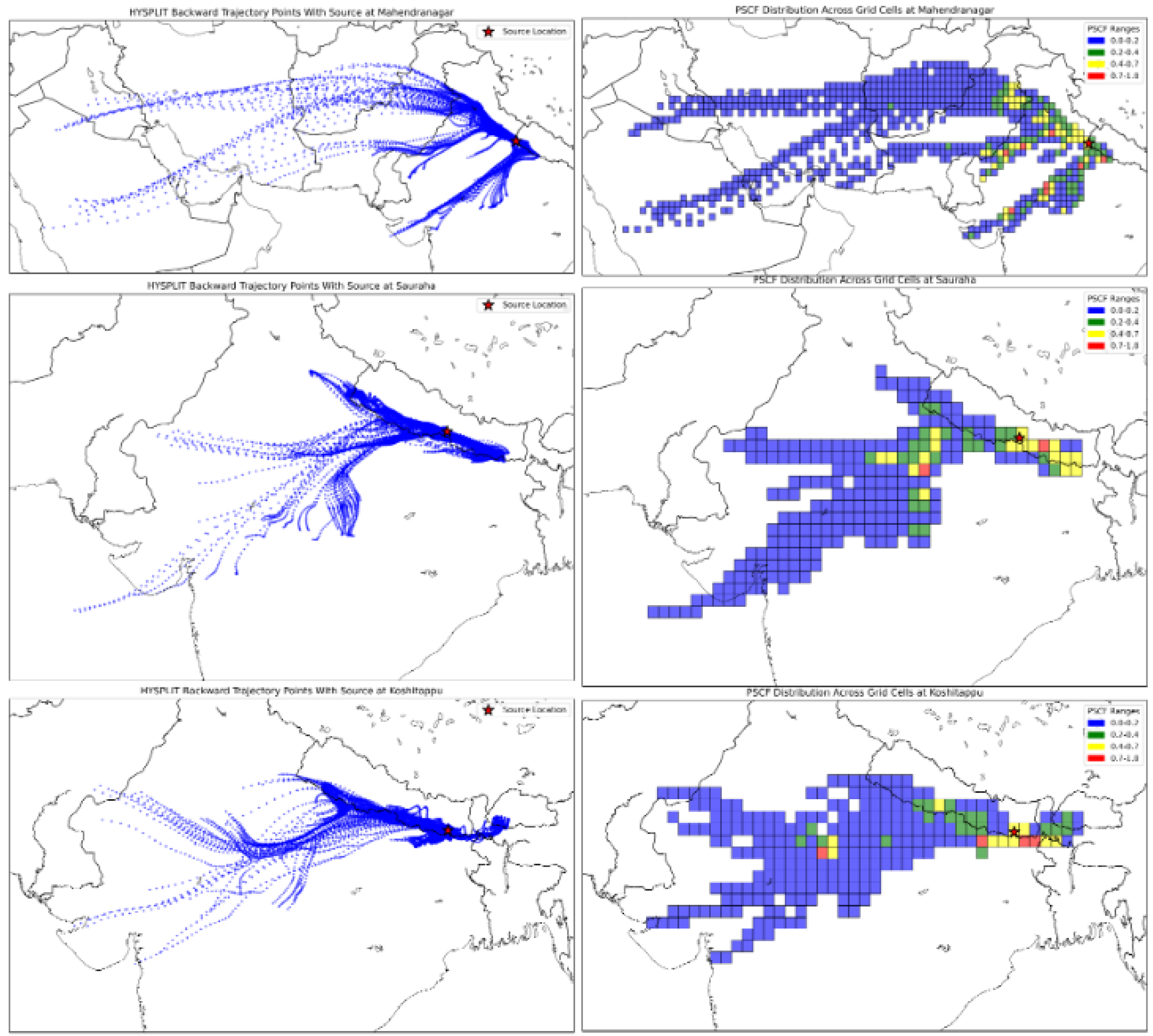

Figure 12: 168-hour HYSPLIT back trajectories and PSCF distribution across grids at 1000m AGL obtained using the GDAS data for the background sites.

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Fire Events

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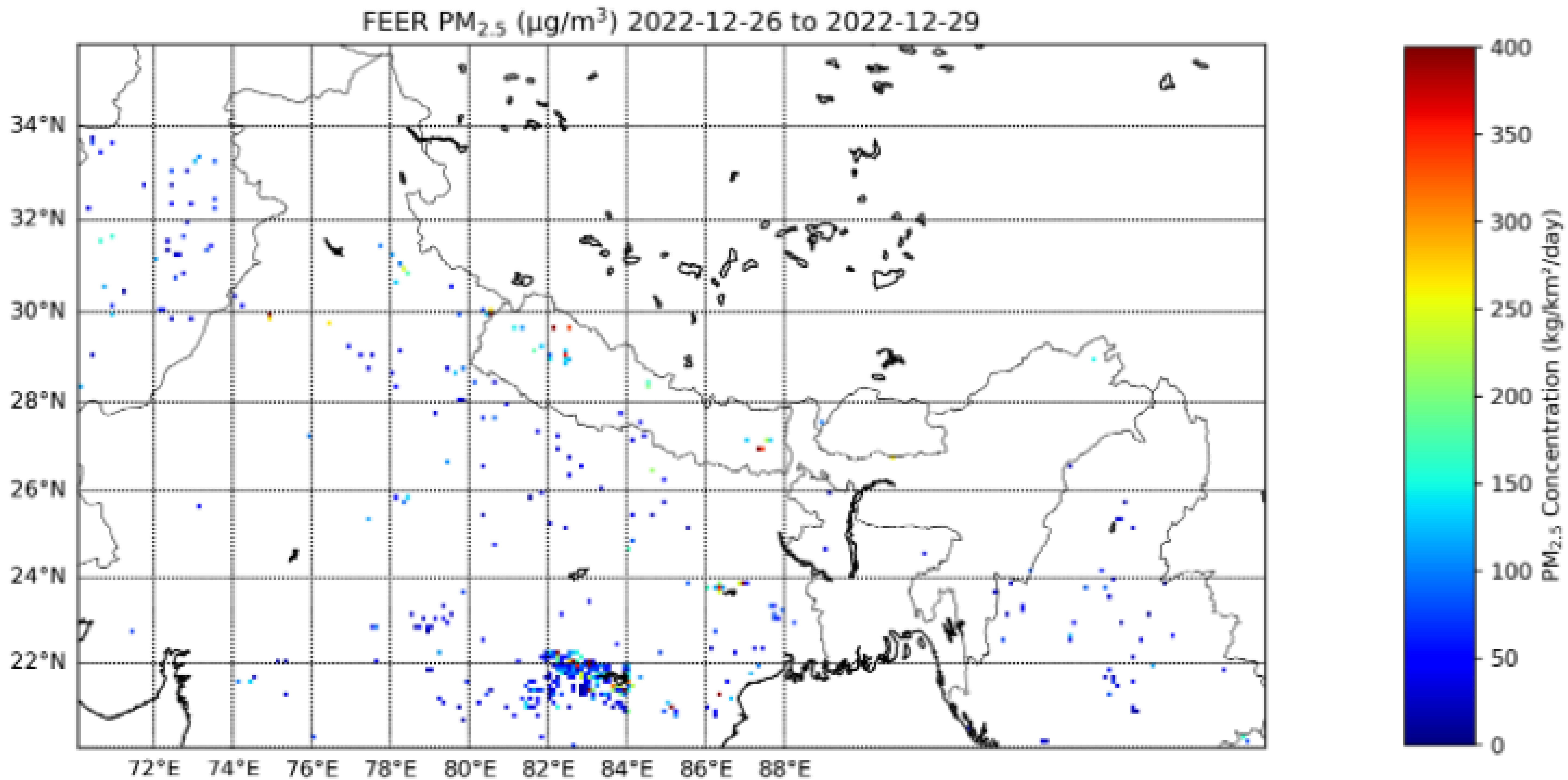

Figure 13: Fire emissions as seen through FEER data around the episodic days.

Conclusions

Atmospheric pollution remains critically hazardous and widely prevalent in developing countries like Nepal. We deployed a network of carefully calibrated low-cost $PM_{2.5}$ sensors across the Terai belt and two major mid-hill valleys in Nepal, uncovering a significant high-pollution episode during December 28-30, 2022, marked by a sudden rise in $PM_{2.5}$ concentrations across all receptor sites. Various meteorological and dispersion analyses, including peak shaving, aerosol optical depth (AOD), air parcel back trajectories, planet boundary layer height (PBLH), ventilation coefficients, and fire events, provided a comprehensive understanding of the complex local and synoptic scale dynamics during this episode. An air mass residence model, Potential Source Contribution Function (PSCF), further demonstrated the role of atmospheric circulation in driving $PM_{2.5}$ levels. Combining these results revealed a strong evidence of cross-border pollution transport through south-easterly winds in eastern Terai and south-westerly winds in western Terai. Our findings highlight the urgent need for transnational environmental stewardship among nations in the Indo-Gangetic Plain (IGP) region.

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